

Water Systems

TABLE 3-9. Source Capacities By District and Island-wide

Installed and Standard Source Capacities By District				Brackish - Manele, Irrigation Grid, Others			
Lana'i City, Koele & Kaunapapa		Manele, Hulopo'e, Palawai, Irrigation Grid		Source		Source	
Source	GPM	Well 2/ Shaft 3*	GPM	Well 1	GPM	Well 9	300
Well 6	550	Well 3**	1,200	Well 12	NIU	Well 14	300
Well 7	NIU	Well 4	900				
Well 8	850	Well 5	NIU				
Maunalei - Shaft 2	NIU						
Tunnels	NIU						
Well 3 (could serve either way)	550/OOS						
Total GPM		Total GPM		Total GPM		Total GPM	
1,400		2,100		940		940	
Total GPD Wells		Total GPD Wells		Total GPD Wells		Total GPD Wells	
2,016,000		3,024,000		1,353,600		1,353,600	
Largest Pump Out		Largest Pump Out		Largest Pump Out		Largest Pump Out	
792,000		1,296,000		864,000		864,000	
Max Day Capacity*		Max Day Capacity *		Max Day Capacity *		Max Day Capacity *	
528,000		864,000		576,000		576,000	
2/3 Installed Less Largest Pump		2/3 Installed Less Largest Pump		2/3 Installed Less Largest Pump		2/3 Installed Less Largest Pump	
Average Day Capacity **		Average Day Capacity **		Average Day Capacity ***		Average Day Capacity ***	
352,000		576,000		384,000		384,000	
2/3 of 2/3 Installed Capacity less largest pump		2/3 of 2/3 Installed Capacity less largest pump		2/3 of 2/3 Installed Capacity less largest pump		2/3 of 2/3 Installed Capacity less largest pump	
Effluent		Manele WWTF		Effluent		Effluent	
Lana'i City WWTF - R-3		500,000		140,000		140,000	
Lana'i City Auxiliary WWTF R-1		400,000					
Total Koele, City and Kaunapapa		2,416,000		Total Manele & IGGP		4,377,600	
Total City, Koele, Kaunapapa Potable		2,016,000		Total Manele & IGGP Brackish		1,353,600	
Total Potable - Standards		352,000		Total Brackish - Standards***		384,000	
				Installed Capacity			
				Total All Island Potable		5,040,000	
				Total All Island Brackish		1,353,600	
				Total All Island Effluent		540,000	
				Total All Island		6,933,600	

* Statewide System Standards indicate that a system should meet max day demand (=Avg Demand x 1.5)

PLUS fire flow in 16 hours pumping, with the largest pump out. Referred to as Max Day Capacity above.

** This means that 2/3 of the above should be able to meet Average Day Demand

*** Standards only apply to brackish water if people or animals depend upon them.

Reservoirs and storage should be sized to meet maximum day demand plus the highest applicable fire flow for a presumed two hour fire. A table of storage is shown in Figure 3-11.

Overall system capacities are summarized below.

Developed & Utilized Resources - System Infrastructure

TABLE 3-10. Summary of System Capacities and Use

Lana'i City, Koele and Related Areas	MGD 2008
Total Installed Capacity	2.416
Installed Capacity of Potable Sources	2.016
Average Potable Use	0.523 metered / 0.605 pumped
Average Non-Potable Use	0.209 to Koele Golf Course
Capacity of Non-Potable Brackish Sources in Use	0.000
Capacity of Reclaimed Water Facilities	0.400
Average Effluent Production	0.235
Potable Storage	2.786
Non Potable Storage	16.8 active / 22.8 total
Approximate Miles of Pipeline	35.59 miles
 Manele, Hulopo'e and Related Areas	 MGD 2008
Total Installed Capacity	4.518
Installed Capacity of Potable Sources in Use	3.024
Average Potable Use	0.375 metered / 0.683 pumped
Average Non-Potable Use - Brackish	0.760 metered / 0.944 pumped
Average Non-Potable Use - Reclaimed	0.073 wtf production
Capacity of Non-Potable Brackish Sources in Use	1.354
Capacity of Reclaimed Water Facilities	0.140
Average Effluent Production	0.073
Potable Storage	2.000
Non Potable Storage	17.85 active / 19.35 total
Approximate Miles of Pipeline	43.04 miles*

(*Estimate does not include roughly 14.81 miles of abandoned or out-of-use pipeline in the Palawai Irrigation Grid)

TABLE 3-11. Lana'i Tanks & Storage

Tanks and Storage							
TANK NAME	TANK CAPACITY IN MG	SPILLWAY ELEVATION	GROUND ELEVATION	YEAR BUILT	TANK MATERIAL	USE	C12 SITE COMMENT
Lana'i City, Koele & Kaunapali							
Maunalei Tank	0.030		1017	1992	Steel	Potable - PWS 237	N Not In Use
Koele Tank	0.734	2057	2025	1995	Bolted Steel w Glass Fused Coating	Potable - PWS 237	N SCADA
Lana'i City - Nihiwai Tank		1878	1830	1951	Steel	Potable - PWS 237	N 2 MG Not In Use
New Lana'i City Tank	2.000	1980	1942	2008	Bolted Steel w Glass Fused Coating	Potable - PWS 237	N Chlorinated at Well 6
Kaunapali Tank	0.022		360	1992	Bolted Steel w Glass Fused Coating	Potable - PWS 237	Y
Lana'i AWWTF R-1 Reservoir							
Lana'i AWWTF R-1 Reservoir	10.000		1518.5			Non-Potable - Effluent	
Koele Clubhouse Lake 1	1.900		1761			Non-Potable - Effluent	Passive
Koele Clubhouse Lake 2	0.900		1741			Non-Potable - Effluent	Passive
Koele 8th Hole Lake/Reservoir	2.000		1739			Non-Potable - Effluent	
Koele 9th Hole Lake/Reservoir	1.100		1746			Non-Potable - Effluent	
Koele 13th Hole Lake/Reservoir	0.900		1991			Non-Potable - Effluent	
Koele 2nd Hole Lake/Reservoir							
Koele 2nd Hole Lake/Reservoir	1.5					Non-Potable - Effluent	
Koele 3rd Hole Lake/Reservoir - Tee	0.900					Non-Potable - Effluent	
Koele 3rd Hole Lake/Reservoir - Green	0.400					Non-Potable - Effluent	
Koele 6th Hole Lake/Reservoir	1.300					Non-Potable - Effluent	
Koele 17th Hole Lake/Reservoir	0.400		1822			Non-Potable - Effluent	
Koele 18th Hole Lake/Reservoir	1.500		1746			Non-Potable - Effluent	
Potable Storage	2.786						
Non Potable Storage	22.800						Including Passive
Non Potable Active Storage	16.800						
Manele, Hulopoe and Portions of Irrigation Grid							
Hii Reservoir	1.000	1823	1810.2		Concrete Lined		N PE Cover
Hii Tank	0.500	1823.5	1791.5	1952	Welded Steel	Potable - PWS 238	N
Manele Breaker Tank 1	0.100	1141	1127	1987	Bolted Steel w Glass Fused Coating	Potable - PWS 238	Y Chlorination Site
Manele Breaker Tank 2	0.100	755.8	742	1987	Bolted Steel w Glass Fused Coating	Potable - PWS 238	N
Manele Breaker Tank 3	0.300	341	326	1987	Bolted Steel w Glass Fused Coating	Potable - PWS 238	N
Wells 9 & 1 Control Tank	0.050	1434.5	1420.5		Steel	Non-Potable - Brackish	Abandoned
Palaw ai Brackish Reservoir	15.000	1211	1239		Lined	Non-Potable - Brackish	

TABLE 3-12. Lana'i Pump Inventory - Source

Well / Source	Pump & Motor Description	Pump Elevation	Calibrated GRM	24 Hrs	16 Hrs	44.44% MAV in GPD	2008 Actual Character
6	Submersible Byron Jackson 23 Stage, 1800 RPM, 2500 V with 200 HP Type H 14" Motor installed 2006	863	550	792,000	528,000	352,000	Potable Drilled 1986 Proposed Potable Drilled 1987 0 Not In Use
7	Submersible Byron Jackson 11 MQH, 20 stage, 1800 RPM with 300 HP Type H, F1 Amp 74 Motor 2300 Volts	782	850	1,224,000	816,000	544,000	Potable Drilled 1990 pump low ered to 783' 09/09 Potable Drilled 1946 Rarely Used Confined Space Issues Potable
2	Vertical Turbine Fairbanks Morse Pomona 3 Stages, 1780 RPM with 75 HP Ingersoll Rand Booster FL Amps 90 voltage 480 Frame-L365TP	1,335	1,200	1,152	1,728,000	1,152,000	768,000 2,418
3	Submersible Byron Jackson #781-S-1808 22 Stage 1800 RPM 23 stage w/Byron Jackson Type H 300 HP 14" Motor Submersible Byron Jackson #841-S-0046, 13MQH, 15 stage, 1800 RPM with Byron Jackson Type H 300 HP 14" Motor Amp 74 2300 Volts	863					Drilled 1950 Out of Service 0 Service Potable Drilled 1950 Out of Service Potable Drilled 1950 Out of Service 0 Service
5	Submersible Crown 340 GRM 9 Stage, 3,470 RPM, with Hitachi 100 HP Motor installed 2005	1,293					
1	Submersible Byron Jackson 16 stage 600-4114-931-R-005 6 MQH 3600 RPM w Franklin Electric 100 HP Motor installed 2005 F1-Amp-148 480 Volt	516	340	300	489,600	326,400	217,600 393,981 Drilled 1945 Brackish
9		466	300	432,000	288,000	192,000	151,440 Drilled 1990 Brackish Drilled 1990 0 Not In Use
12		-5					
14	Submersible Byron Jackson Hitachi 125 HP Motor installed 2003	400	300	432,000	288,000	192,000	404,714 Drilled 1995 Brackish
Source Capacity				4,440	6,393,600	4,262,400	2,841,600 2,241,222

Water Systems

TABLE 3-13. Lanai Pump Inventory - Boosters and Totals

Boosters	Description	Quantity	Unit Cost	Total Cost	Status	Source	Out of Service
Maunaloa	Byron Jackson Can Type Vertical Booster 111LQ 3600 RPM 6 Stages with Vertical Solid Shaft 20 HP 3600 RPM 444 VP Frame WP-1 Enclosure 225 Amp F1	481				Potable Booster Quasi-Source	0
	Well 2 / Shaft Ingersoll Rand Booster Pump 3 Booster 40 HP					Potable Source Rarely Used	
	Manele SPS A - 2 Pumps	295 ea.	424,800	283,200	188,800	Effluent Booster	
	Manele SPS B - 2 Pumps	490 ea.	705,600	470,400	313,600	Effluent Booster	
	Manele SPS # 1 - 2 Pumps	550 ea.	792,000	528,000	352,000	Effluent Booster	
	Manele SPS # 2 - 2 Pumps	550 ea.	792,000	528,000	352,000	Effluent Booster	
	Manele SPS # 3 - 2 Pumps	550 ea.	792,000	528,000	352,000	Effluent Booster	
	Koele WW - Green 4 40 HP	1,992	280	403,200	268,800	179,200	
	Koele WW - Green 17 40 HP	1,748	800	280	403,200	268,800	179,200
		4,335	4,312,800	2,875,200	1,916,800	0	
		8,775	10,706,400	7,137,600	4,758,400	2,241,222	
Subtotal Potable in Use		3,500	5,040,000	3,360,000	2,240,000	1,291,087	
Subtotal Brackish in Use		640	921,600	614,400	409,600	545,421	
Subtotal Wastewater in Use		3,535	3,909,600	2,606,400	1,737,600		

Lana'i City Water System - Potable

The Lana'i City Water System serves Koele, Lana'i City and Kaunapali. The system has roughly 1,400 service connections, served by two wells, three tanks and roughly thirty-five miles of potable line. Source for this system is currently drawn from two active wells, Well 6 (aka Kaiholena Well 6 - USGS #5054-01) at 1,910' and Well 8 (USGS # 4753-01) at 1,902'. Well 3 was once an important source for this system, but has since been taken out of service. A replacement for this well is in progress as of this draft, and scheduled to be on-line in 2010.

The system is untreated with the exception of the standard required chlorination, which takes place at the sources, and again at Kaunapali Harbor tank. Koele, Lana'i City and Kaunapali represent three service zones on the system.

Koele is served by Wells 3 and 8, via the 734,000 gallon Koele Tank, with a spillway at 2057.5'. The low elevation limit of this pressure zone is about 1,740'. The Koele Tank primarily serves Koele Villas and lots and the Lodge at Koele, but water from this tank can drop to the City through a PRV.

Lana'i City is served primarily by Well 6. Well 6 feeds directly to the New Lanai City Tank, with a spillway elevation of 1,980 feet. Water from Wells 3 (once replaced) and 8 can also contribute source to Lana'i City via a PRV from the Koele service area. Well 6 feeds directly to the New Lana'i City 2,000,000 gallon tank, with a spillway elevation of 1980'.

Kaunapali is fed from Lana'i City via a 2-1/4" pipe to the 22,400 gallon steel storage tank at Kaunapali, with a spillway elevation of 375'. This tank services Kaunapali Harbor and small surrounding developments.

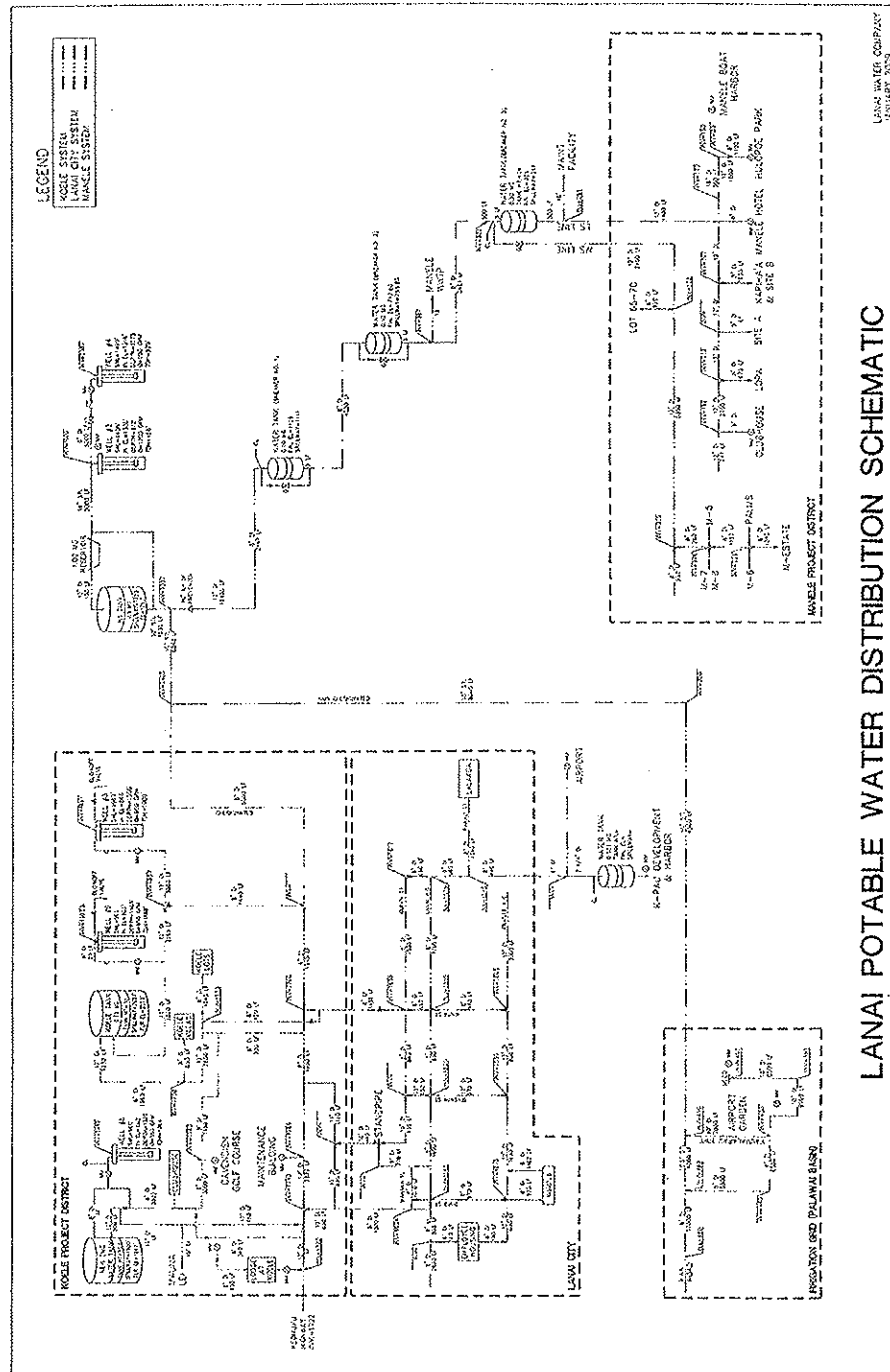
Lanai City - Non Potable - Reclaimed Water

Two wastewater treatment plants serve Lana'i City. The County's Lana'i City Wastewater Treatment Facility has a capacity of about 500,000 gallons per day and treats water to R-3 quality. In calendar year 2008, the Lana'i City Wastewater Treatment Facility had an influent of about 308,412 gallons per day, and produced about 245,456 GPD of effluent.

From the Lana'i City Wastewater Treatment Facility, effluent proceeds to the Lana'i City Auxiliary Treatment Facility where it is further treated to R-1 quality water. The Auxiliary Treatment Facility has a capacity of about 400,000 GPD. In 2008, with an influent of 245,456 GPD, the Auxiliary Treatment Facility produced about 234,093 GPD of R-2 water.

The Auxiliary Treatment Facility has a storage capacity of about 10 MG, with additional storage in water features at the "Experience At Koele" Golf Course of about 13.1 MG. The non-potable system has roughly three miles of waterline. About 209,721 gallons per day were pumped to the "Experience At Koele" Golf Course from the Auxiliary Water Treatment Facility during 2008.

FIGURE 3-18. Lana'i Potable Water System Schematic



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Manele Water System - Potable

The Manele Water System serves Manele Resort, Hulopo'e Beach Park, and the Manele small boat harbor, as well as the Palawai Irrigation Grid. It has roughly 200 service connections and is served by two wells, five tanks and roughly thirty-five miles of potable or non-chlorinated waterlines. Water for the Manele system is drawn primarily from Well 4 (aka Soule's Bench Well, USGS #4952-02), with very occasional withdrawals from Well 2/Shaft 3 (USGS # 4953-01). From Well 4, at an elevation of about 2,327', it proceeds to the 1 MG concrete Hii Reservoir and 0.5 MG Hii Tank at 1,823'. From Hii, water is carried to Manele via three steel breaker tanks in series at spillway elevations of 1,141', 755.8' and 341' respectively. Breaker Tanks 1 and 2 have capacities of 100,000 gallons each. Breaker Tank 3 has a capacity of 300,000 gallons. Water for the Palms and multi-family estates at the west end of Manele is channeled into a line just above Breaker Tank 3. From Breaker Tank 3, water continues to the Harbor, the Beach Park and the Hotel.

Manele Water System - Non Potable**Brackish**

Brackish water for landscaping at Manele comes from Wells 1 (USGS # 4853-02), 9 (USGS # 4854-01) and 14 (USGS # 4854-02) at 1,265', 1,411' and 1,193' respectively. From Well 9 water enters a 0.5 MG control tank at 1,420.5' with a spillway elevation of 1,434.5'. From this tank and the other wells, water proceeds to the 15 MG Reservoir with a spillway elevation of 1,211'. Water is then piped via two 40,000 gallon breaker tanks with spillway elevations of 1,000' and 712' toward Manele. Just above Manele, brackish water is blended with reclaimed effluent for golf course irrigation. There are roughly seven miles of brackish waterline.

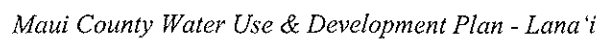
Reclaimed Water

The Manele Wastewater Treatment Facility has a capacity of 140,000 GPD. During calendar year 2008, with an influent of 77,281 GPD, it produced 72,940 GPD of effluent. From the effluent reservoir, this water is pumped directly to the Manele Golf Course via roughly one mile of wastewater line.

TABLE 3-14. Wastewater Facility Capacity, Influent and Effluent on Lana'i - 2008

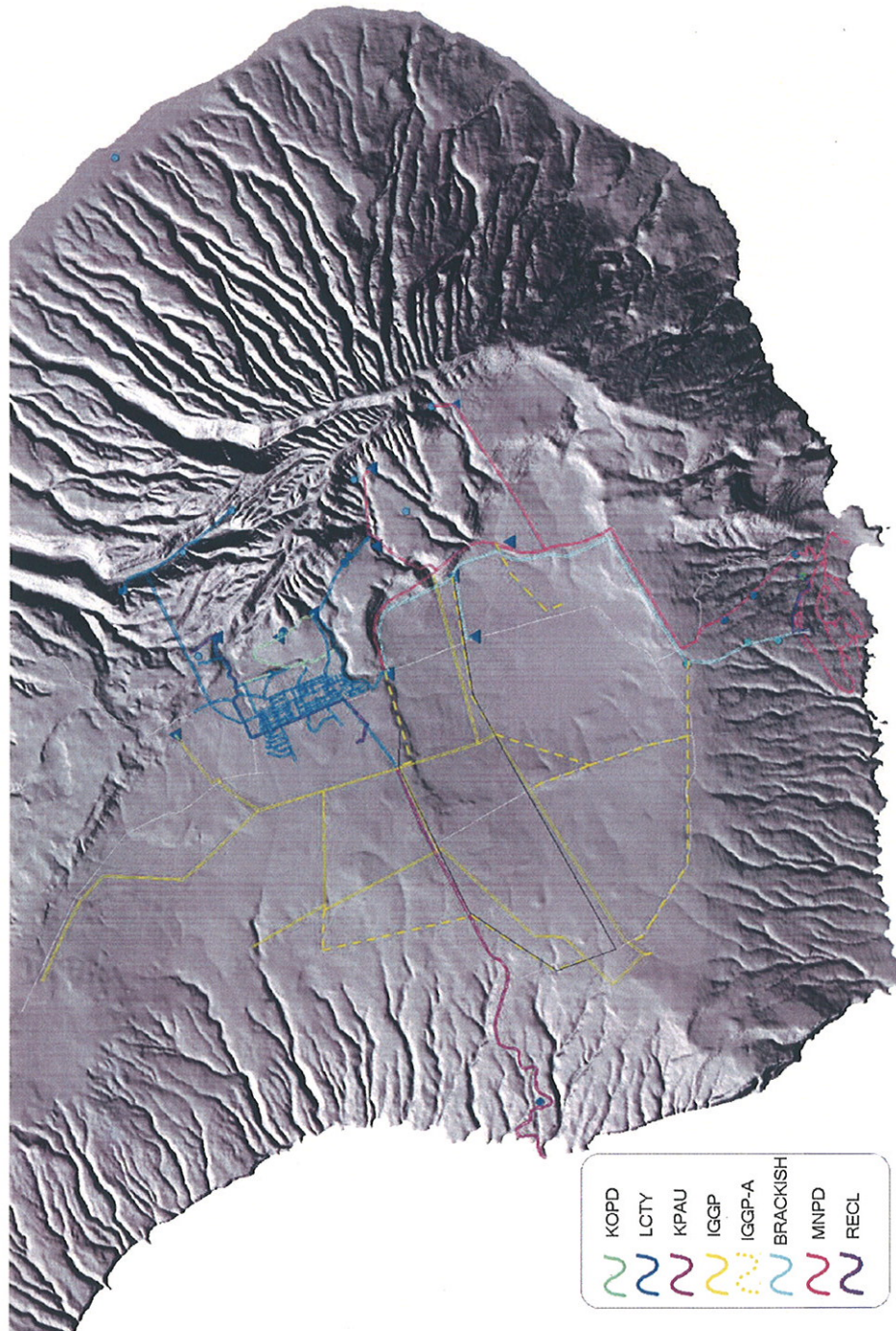
Name		Capacity	Average Influent	Average Production
Lana'i City WWTF	R-3	500,000	308,412	
Lana'i City Auxilliary WWTF	R-1	400,000	245,456	234,093
Manele WWTF	R-1	140,000	77,281	72,940

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Water Systems

FIGURE 3-21. Lana'i Water Systems By District



Pipelines

In all, Lana‘i has roughly 93.44 miles of pipeline, as measured from GIS plots. Of this, about 78.63 miles are active, and 14.81 miles are abandoned or out of use. The age and condition of some of these lines, combined with the lack of customer base to generate adequate revenues for necessary replacements, is a matter of serious concern to both Lana‘i Water Company, Inc. and the community it serves. Long segments of pipe in the irrigation grid, and to the west, south and east of Lana‘i City, are in need of repair, replacement or in some cases possibly abandonment. A portion of the line from Hi‘i tank down to the Palawai Basin is unburied and in a fire-prone area. In addition, some lines are either made of materials or too small in diameter to satisfy system standards. These situations will be a challenge for the utility in the coming decade.

FIGURE 3-22. Palawai Grid Pipe Age Data Dotted lines in this image are abandoned.

